

SHRAVANKUMAR NAGARAJAN

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PROFESSIONAL SUMMARY

Lead Full Stack Developer with senior-level expertise in **Java, Spring Boot, Python, and Node/React**. Proven track record designing high-throughput data pipelines, automating **CI/CD** workflows, and building scalable **microservices** and **monitoring** tools on **UNIX/Linux** platforms. Skilled in both **SQL** and **NoSQL** databases. Currently focused on delivering innovative solutions with **AI/ML**, including **LLMs** and **Agentic AI**.

EDUCATION

San Jose State University

Jan 2025 - Dec 2026

Master's, Applied Data Intelligence (GPA: 3.78)

- **Coursework:** Machine Learning, Deep Learning, Generative AI, Agentic AI, Big Data Algorithms, Distributed System, Data Warehouse, Analysis

SKILLS

- **Programming Languages:** Python, Java, JavaScript, Node, SQL, Bash, Scala, C/C++, Rust
- **AI & GenAI:** LangChain, LangGraph, Spring AI, LlamaIndex, AutoGen, LLaMA, Ollama, RAG Pipelines, Agentic AI, Hugging Face
- **Frameworks & DevOps:** Spring/Spring Boot, Project Reactor, FastAPI, ExpressJS, Next.js, Nest.js, Redux, React, UNIX/Linux, Docker, Kubernetes, Helm, Terraform, Concourse CI, Jenkins, Ansible, AWS, Azure, GPU, Grafana, Prometheus, ELK, New Relic, Open Telemetry
- **ML/DL:** Scikit-learn, PyTorch, Computer Vision, Random Forest, LSTM, autoencoders, encoder/decoder, VAE, CNN/RNN, XGBoost
- **Data Engineering & Streaming:** PySpark, Airflow, Kafka, AWS Glue, Databricks, Trino, Flink, Apache Camel, Apache Pulsar, Pandas
- **Databases & Streaming:** MySQL, Elasticsearch, MongoDB, Cassandra, PostgreSQL, ChromaDB, FAISS, Redshift, Big Query, Neo4j, Kafka
- **Mathematical Foundations:** Linear Algebra, Probability, Statistics, Optimization, Hypothesis Testing, Gradient Descent
- **Soft Skills & Extras:** Curiosity, Quick Learner, Git, Jira, Confluence, Agile, Research Skills, Information Gathering

PROFESSIONAL EXPERIENCE

The Bank of New York Mellon | *Lead Full Stack Developer*

Sep 2022 - Jan 2025

- Created **micro-frontends** and **middleware** with **React, GraphQL, Next.js, and Express.js** to automate cloud provisioning, aggregate infrastructure metrics, trigger **Bash/Ansible/Terraform** workflows, and monitor and restart automation pipelines improved provisioning speed.
- Architected microservices with **Spring Boot, Spring Data, and Spring Cloud (Eureka, Config Server, Consul)** and automated **CI/CD** and **infrastructure** provisioning using **Ansible** and **HashiCorp** tools like **Packer** and **Terraform**, accelerating release cycles.
- Contributed to **R&D design** of a **multi-agent** autonomous operations system, comprising monitoring, diagnosis, remediation, and escalation agents that detected provisioning failures, **infrastructure** workflows, and enforced **responsible AI** by **policy-driven human-in-the-loop** controls.
- Developed and deployed **XGBoost, Autoencoder, and LSTM** models using **PyTorch** and **scikit-learn** for **capacity forecasting** and **anomaly detection**, reducing false alerts by **35%** and delivered CI/CD with **ArgoCD, Helm, and AWS EKS**.

Elsevier | *Senior Software Engineer*

Jul 2020 - Sep 2022

- Ingested **12TB** of scholarly content from **S3** using **PySpark** with **Groovy, Kafka, and Kafka Connect** into **Elasticsearch**, enabling **sub-500 ms** semantic search across **10M** documents per day for millions of **Scopus** and **Mendeley** researchers.
- Integrated **open source monitoring** and **testing frameworks** to harden a **star schema** in **AWS Redshift data warehouse**, improving data quality/reliability, and compliance while streamlining cloud infrastructure and reducing operational overhead, **saving \$250K annually**.
- Created a **reviewer-recommender system** that ranks peer reviewers with weighted scoring, built with **FastAPI, MongoDB, and PostgreSQL**, and deployed via **Jenkins** pipelines, which accelerated reviewer assignment and improved match quality for submissions.

Accenture | *Senior Software Engineer*

Aug 2019 - Jul 2020

- Produced **15+ microservices** and **5+ micro-frontends** for network monitoring using **Scala, Shell Scripts, React, and Kafka**, with **end-to-end CI/CD** via **Concourse**, reducing deployment time by **1 hour weekly** and improving availability to **four nines of uptime**.
- Built **graph-based services** on **DataStax Enterprise Graph** and **Cassandra** processing **millions of events/day**, integrating **Prometheus, Grafana, and Alerta** to enable **real-time telemetry**, cutting **MTTR** by **35%** and reducing false alerts by **25%**.

Cognizant | *Associate*

Mar 2018 - Aug 2019

- Delivered insurance suite with **Guidewire InsuranceNow** and built UI modules with **jQuery** and **Angular**, led migration from **Maven** to **Gradle**, and created a metrics framework to guide feature investments and speed up enhancement identification.

Tata Consultancy Services | *Systems Engineer*

May 2015 - Mar 2018

- Migrated a large retail platform from **SOAP** to **REST APIs**, cutting **99th-percentile** latency to under **200 ms** and **doubling throughput** through optimized **Oracle PL/SQL** queries and distributed caching with **Terracotta**.
- Improved **batch** and **event processing** speed by adding **multithreading** and **distributed microservices** with **Spring Batch Integration**, which reduced job completion times, saving **ten hours** of processing time per month.

Projects

Kayak-Clone AI-Powered Travel Search/Booking Agent | [GitHub](#) | *San Jose State University*

Oct 2025 - Dec 2026

- Built a **multi-agent AI concierge service** integrated into the travel platform that decomposed natural-language user requests (e.g., flexible itinerary preferences) into domain tasks (flight, hotel, car), orchestrating intent parsing, and retrieval to deliver personalized recommendations.
- Built **RAG workflows** with **ChromaDB embeddings** and **LLMs**, benchmarked with tools such as **Hugging Face Evaluate** and **LangSmith**, achieving **92%** recall in automated precision/recall testing of retrieval and reasoning quality.
- Experimented with multiple LLM/diffused models (**Phi-4, Mistral, Gemma 3, and OpenAI APIs**) and benchmarked conversational accuracy with automated test suites (**Hugging Face Evaluate + custom scenarios**), validating **90% precision** and cost-aware deployment observability.